ZCAS University

CIT 3341 E-COMMERCE AND SECURITY

FINAL EXAMINATION

4TH JUNE, 2025

14:00 -17:00 HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

ZamBuy, a Zambian e-commerce startup, was launched two years ago to provide a digital marketplace for local farmers, craftsmen, and small-scale retailers to sell their products across the country. The platform offers online payment options using mobile money (e.g., Airtel Money, MTN MoMo), credit/debit cards, and bank transfers. It also partnered with a local logistics company for delivery services.

Despite growing interest in online shopping, ZamBuy faces **low adoption of its online payment systems**, especially among rural users and first-time buyers.

Required:

a) Identify five possible reasons for the low adoption of online payment systems on ZamBuy. (10 marks)

b) For each reason you identified, suggest a practical solution that could be applied in Zambia to encourage more users to use online payments. (10 marks)

c) As a consultant, propose four specific technical or design improvements ZamBuy could implement on its platform to improve the user experience and encourage more secure online transactions. (12 marks)

d) ZamBuy recently experienced an incident where some customers received phishing emails that looked like official ZamBuy communications and were tricked into entering their login credentials on a fake website.

   i. Identify **two security risks** that this phishing incident exposes for ZamBuy and its users. (4 marks)

   ii. Propose **two appropriate security controls or strategies** that ZamBuy could implement to prevent such incidents in the future. (4 marks)

[Total: 40 marks]

SECTION B: Attempt THREE questions in this section

Question 2

a) As a consultant, you have been hired by a traditional brick-and-mortar retailer who is contemplating the expansion into e-commerce. In your analysis, discuss separately the

benefits and disadvantages of e-commerce for the merchant, the consumer, and society as a whole. (12 marks)

b) Explain the legal, social and ethical issues in the e-commerce environment. (8 marks)

[Total: 20 marks]

Question 3

Identify the type of e-commerce in the following scenarios. Explain the reason for your choice. Each scenario carries two marks (2 marks).

i. Mukwala is a pharmaceutical distributor who wants to establish an e-commerce platform to sell medicines directly to hospitals. Suggest the type of e-commerce that best fits the given scenario.

ii. Ngozi is a supplier of office furniture and wants to expand her business by establishing an online presence. She intends to sell her products directly to businesses and organizations. Suggest the type of e-commerce that would be most suitable for Ngozi's business in this scenario.

iii. Kwame is a farmer who wants to sell his organic produce directly to restaurants and hotels in the area. He plans to set up an online platform to facilitate these transactions.

iv. Lilian runs a graphic design agency and wants to offer her services to individual clients and businesses through an online platform.

v. Omar is a wholesaler of electronics and wants to leverage the power of e-commerce to reach retailers and other businesses. He is considering various options for establishing an online presence.

vi. Jane has a collection of vintage clothing and accessories that she wants to sell to individual consumers online.

vii. Amara is an artist who wants to sell her artwork directly to consumers through an online platform.

viii. TechSupplies is an IT equipment provider that wants to sell its products to government agencies and departments.

ix. GlobalConsult is a consulting firm that specializes in providing advisory services to government entities. They want to establish an online platform to streamline their interactions with government clients.

x. The Ministry of Education in Zambia wants to establish an e-commerce platform to facilitate procurement and exchange of educational resources with other government entities from different countries.

[Total: 20 marks]

Question 4

a) Differentiate the following terms

i. Asymmetric cryptography and symmetric cryptography (5 marks)

ii. Cipher text and message digest (5 marks)

iii. SSL/TLS and SET security Protocol (5 marks)

iv. Digital Envelope and Digital signature (5 marks)

[Total: 20 marks]

Question 5

a) Briefly explain the importance of the following security requirements in the context of e-commerce:

i. Confidentiality

ii. Integrity

iii. Availability

iv. Authenticity

v. Non-repudiation

vi. Privacy (6 mark)

b) Using the SmartBuy scenario, **discuss** how each of the following **cryptographic primitives** contributes to fulfilling the e-commerce security requirements listed in part (a). Provide relevant examples from the scenario.

i. **Encryption** (5 marks)

ii. **Hashing** (4 marks)

iii. **Digital signature** (5 marks)

For each, clearly indicate:

- Which security requirements it addresses
- How it achieves those goals using the scenario context

[Total: 20 marks]

END OF EXAMINATION